

SILENT FAILURE

Who tells you when your ML model breaks?

A model trained in a notebook works great. The moment it goes live, nobody knows if it still does.

INDUSTRY ESTIMATE

60%

of ML models degrade silently within 3 months of deployment.

THE LIFECYCLE GAP

01

Notebook

Train, validate, celebrate.

02

Production

Serve real, messy traffic.

?

Monitoring

Missing. Customers notice first.

THE STACK

DistilBERT, served on a free CPU.

40% smaller, 60% faster than full BERT, with under 3% accuracy loss. The right trade-off when there is no GPU.

**SINGLE
WORKER**

The model is a singleton. Multiple workers would each load 270 MB into RAM and exhaust the container.

01

FastAPI + DistilBERT

distilbert-base-uncased-finetuned-sst-2-english

02

Docker multi-stage build

Model weights baked in. No cold-start download.

03

HuggingFace Spaces hosting

Free HTTPS, auto-redeploy via CI.

04

GitHub Actions CI

lint → test → build → push → deploy.

05

Retrain gate

Bad data PRs auto-rejected with an explanation.

FOUR LAYERS OF DETECTION

Live API · sahil147-sentiment-api.hf.space

GET /docs · /predict · /drift

 /docs

Sentiment API

FastAPI · OpenAPI 3.1

POST/predict

GET/drift

GET/health

Three endpoints. One source of truth.

 GET /drift

```
{
  "status": "alert",
  "window": 100,
  "alerts": [
    "non_ascii_frac"
  ],
  "non_ascii_frac": 0.91,
  "vocab_novelty": 0.42
}
```

Spanish surge caught before a complaint lands.

 POST /predict

```
{
  "label": "POSITIVE",
  "score": 0.9987,
  "latency_ms": 142,
  "model_version":
    "a3f9c21"
}
```

model_version traces every prediction back to the exact git commit.

LIVE IN PRODUCTION

Shipping. Measured. Holding.

/PREDICT LATENCY

130–200ms

CPU only · no GPU

CI PIPELINE

~3min

lint · test · build · push

TEST SUITE

Allgreen

0 ruff warnings



Drift alert fires correctly

Tested with 100 non-English requests. Alert raised.



Retrain gate blocks bad data

Corrupted training set rejected with an explanation.

ROADMAP

Three honest next steps.

Move from grep-able logs to dashboards, persistent state, and safe rollouts.

TODAY

Live, monitored, and reproducible.

01 /metrics endpoint

Prometheus format. Grafana dashboards for request rate, latency percentiles, drift signals.

02 Redis-backed drift storage

Signals survive restarts and aggregate across workers. The window stops resetting.

03 Shadow deployment

Route 5% of live traffic to a candidate model. Compare predictions before promoting.